3.46 Consider following two nonminimum phase systems.

$$G_1(s) = -\frac{2(s-1)}{(s+1)(s+2)} \qquad G_2(s) = \frac{3(s-1)(s-2)}{(s+1)(s+2)(s+3)}$$

a.) Sketch unit step response for $G_1(s)$ & $G_2(s)$, paying close attention to the transient part of response
b.) Explain difference in behavior of two responses as it relates to the zeros

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* step: $\mathcal{L}\{u(t)\} = \frac{1}{s}$

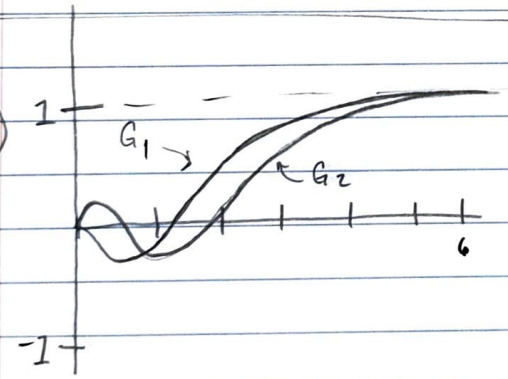
$$G_1(s)\cdot\frac{1}{s} = \frac{-2(s-1)}{s(s+1)(s+2)} = \frac{1}{s} - \frac{4}{s+1} + \frac{3}{s+2}$$

$$g_1(t) = [1 - 4e^{-t} + 3e^{-2t}]u(t)$$

$$G_2(s)\cdot\frac{1}{s} = \frac{3(s-1)(s-2)}{s(s+1)(s+2)(s+3)} = \frac{1}{s} - \frac{9}{s+1} + \frac{18}{s+2} - \frac{10}{s+3}$$

$$g_2(t) = [1 - 9e^{-t} + 18e^{-2t} - 10e^{-3t}]u(t)$$

a.)



b.) G_1 = odd # of zeros = undershoot
G_2 = even # of zeros = overshoot

3.53 Use Routh's stability criterion to determine how many roots w/ + real parts.

a.) $s^4 + 8s^3 + 32s^2 + 80s + 100 = 0$

Routh array:

s^4: 1 32 100

s^3: 8 80

s^2: $\frac{(8)(32)-(1)(80)}{8} = 22$ $\frac{8(100)-(1)(0)}{8} = 100$

s^1: $\frac{(22)(80)-8(100)}{22} = 43.63$

s^0: 100

No poles in RHP

0 roots with pos real parts

3.55 TF: $KG(s) = \frac{K(s+4)}{s[(s+0.5)(s+1)(s^2+0.4s+4)]}$ 6/6

Using Routh's stability criterion, determine range of K for which system is stable when $1+KG=0$.

Char eq: $1+KG(s)=0$

$$1+\frac{K(s+4)}{s[(s+0.5)(s+1)(s^2+0.4s+4)]}=0$$

$$s[(s+0.5)(s+1)(s^2+0.4s+4)]+K(s+4)=0$$

$$s^5+1.9s^4+5.1s^3+6.2s^2+2s+Ks+4K=0$$

$$s^5+1.9s^4+5.1s^3+6.2s^2+(2+K)s+4K=0$$

Routh array:

s^5:	1	5.1	$(2+K)$	← (odd powers)
s^4:	1.9	6.2	$4K$	← (even powers)
s^3:	1.837	$2-1.105K$		
s^2:	$4.13+1.14K$	$4K$		
s^1:	$\frac{-1.26K^2-9.63K+8.26}{4.13+1.14K}$			
s^0:	$4K$			

$\frac{1.9(2+K)-4K}{1.9} = 2-1.105K$

$\frac{(1.837)(6.2)-1.9(2-1.105K)}{1.837}$

$\frac{11.389-3.8+2.099K}{1.837}$

$4.131+1.143K$

$\frac{1.837(4K)-1.9(0)}{1.837}=4K$

$\frac{(4.13+1.14K)(2-1.105K)+(1.837)(4K)}{4.13+1.14K}$

$\frac{-1.26K^2-9.63K+8.26}{4.13+1.14K}$

every element in 1st column must be >0

s^0: $K>0$
s^1: $K<0.78$
s^2: $K>-3.62$
s^3: $K<1.81$

$\boxed{0<K<0.78}$